

JEEVAANANTHAN M

FULL-STACK ENGINEER | AI & DATA ENGINEERING

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PROFILE SUMMARY

Full-stack engineer with about 3 years of experience building and deploying production MERN applications for healthcare, CRM, and e-commerce. Most of my work is on the backend: REST APIs in Node.js and Express.js, MongoDB and MySQL schema design, authentication, and deployment on Linux servers. On top of that I build AI applications and data pipelines: ZenithDesk, a multi-tenant support platform with an AI chatbot, now live in production; Eventa, an open-source offline AI assistant for Node.js developers with its own fine-tuned model, published on npm; and an e-commerce sales data pipeline with Python, Apache Airflow, and dbt.

EXPERIENCE

MERN STACK DEVELOPER | PENTABAY SOFTWARES | CHENNAI || MAY-2025 – PRESENT |

- **Front-End Development:** Built dynamic, responsive user interfaces using React.js, focusing on cross-browser compatibility and optimized performance.
- **Component Reusability:** Developed maintainable and scalable front-end components leveraging React hooks and the Context API for efficient state management.
- **Back-End Development:** Built robust server-side applications with Node.js and Express.js, implementing CRUD operations and RESTful APIs.
- **Middleware Integration:** Developed middleware for request validation, error handling, and data security to ensure reliable system operations.
- **Database Management:** Managed MongoDB databases using Mongoose ORM for schema definition, data validation, and streamlined operations.
- **Query Optimization:** Implemented indexing strategies to enhance query performance and system scalability.
- **Routing and Navigation:** Integrated React Router for seamless navigation and dynamic routing in web applications.
- **Performance Optimization:** Enhanced application efficiency by optimizing component rendering and load times.
- **Deployment and Hosting:** Deployed full-stack applications to production environments, managing build processes, environment configurations, and server setup to ensure high availability and reliability.

MERN STACK DEVELOPER | FACES SYNC | CHENNAI || OCT-2023 – MAY-2025 |

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MEDICAL BILLING SPECIALIST | KMCH | COIMBATORE || 2020 – 2021 |

Medical Billing:

- Processed and submitted insurance claims accurately and efficiently to ensure timely reimbursements.
- Conducted thoroughly verified patient information, insurance coverage, and billing codes.
- Resolved discrepancies in billing and claims by collaborating with insurance providers and internal departments.
- Ensured compliance with medical billing regulations and hospital policies.
- Maintained meticulous records of billing activities, payment status, and financial reports.
- Assisted patients with billing queries, payment plans, and insurance processing.

EDUCATION

HINDUSTAN COLLEGE OF ARTS AND SCIENCE | COIMBATORE | | 2015 – 2018 |

Bachelor in Commerce with Corporate Secretaryship

PROJECT

ZENITHDESK – MULTI-TENANT CUSTOMER SUPPORT PLATFORM | PERSONAL PROJECT |

System Technology Used: React.js, Vite, Tailwind CSS, Node.js, Express.js, MySQL (Knex), JWT, Playwright, GitHub Actions

Live: portal.zenithdesk.site | GitHub: github.com/Jeeva1398/zenithDesk

BACKEND DEVELOPMENT:

- Built the REST API with Node.js, Express.js, and MySQL (Knex) for a multi-tenant help desk, where each company's data is kept separate by `org_id`.
- Added a tenancy guard that checks every SQL query, so data from one company cannot leak into another.
- Set up JWT login for agents, customers, and admins with short-lived access tokens, rotating refresh tokens, and email OTP sign-in for customers.
- Added rate limiting, a CORS allowlist, Helmet headers, and file type checks on uploads.

DATA AND REPORTING:

- Designed a star-schema reporting database in MySQL (organization, agent, and category dimensions with a daily ticket fact table).
- Wrote an incremental ETL job in Node.js and SQL that runs on a cron schedule and feeds the analytics dashboard.

KEY FEATURES:

- Tickets with comments, attachments, saved views, macros, SLA targets, and global search.
- Knowledge base, customer self-service portal, and a super-admin area across all companies.
- Around 155 Playwright tests (API and UI) that run in GitHub Actions on every push.

ZENITHDESK AI CHATBOT WIDGET | PERSONAL PROJECT |

System Technology Used: React.js, Shadow DOM, Vite, Node.js, Express.js, Groq, Ollama, Zod, SQLite

Live: iamjeeva.in (chat widget on the site) | GitHub: github.com/Jeeva1398/zenithDesk-chat

FRONTEND INTEGRATION:

- Built a chat widget in React that runs inside a Shadow DOM, so it can be added to any website with one script tag without breaking the site's styles.

BACKEND DEVELOPMENT:

- Built the Express.js server that uses Groq-hosted models, with a local Ollama model as a fallback.
- Used Zod to validate the details the model pulls from a chat, with a retry and a rule-based fallback when the output is not valid.
- Connected the widget to the ZenithDesk API to create real tickets, enquiries, and attachments.

KEY FEATURES:

- Answers questions from each company's knowledge base and shows the articles it used.
- Customers can check their ticket status after verifying with email and OTP.
- One deployment serves many companies, each identified by its own widget key.

EVENTA – OFFLINE AI ASSISTANT FOR NODE.JS DEVELOPERS | OPEN-SOURCE PROJECT |

System Technology Used: TypeScript, Node.js, node-llama-cpp, Ollama, Qwen2.5-Coder, Python (Unsloth QLoRA), Vitest, GitHub Actions

npm: npmjs.com/package/@jeeva1398/eventa | GitHub: github.com/Jeeva1398/eventa | Model:

huggingface.co/jeeva1398/eventa-1.5b-gguf

CLI DEVELOPMENT:

- Built a TypeScript CLI that explains Node.js crashes and TypeScript errors using the project's own source code, reviews git diffs, and audits npm dependencies, all offline with no API key.
- Wrote static checks (missing await, SQL and command injection, empty catch, disabled TLS) that guide the model's code review.
- Made the dependency audit deterministic, with exact fix commands and verified breaking-change notes, so the model cannot invent versions.

MODEL TRAINING:

- Fine-tuned Qwen2.5-Coder-1.5B with QLoRA on a free Kaggle GPU and published it on Hugging Face as a 1 GB GGUF model.
- Generated the training data from real crashing programs; on held-out tests, review precision went from 48% to 100% and answers became about 2x faster.

KEY FEATURES:

- GitHub Action that reviews every pull request on the CI runner and posts one comment.
- Pinned, checksum-verified model and runtime downloads; CI on Linux, Windows, and macOS; automated npm releases.

E-COMMERCE SALES DATA PIPELINE | PERSONAL PROJECT |

System Technology Used: Python, MySQL, PostgreSQL, dbt, Apache Airflow, Docker, GitHub Actions, Supabase, Streamlit, Metabase

Live: jeeva-ecom-sales.streamlit.app | GitHub: github.com/Jeeva1398/E-Commerce-Sales

DATA PIPELINE:

- Built a batch ELT pipeline that extracts a sales database from MySQL with Python and loads it into a Postgres warehouse with COPY, in one transaction so a failed run keeps the previous load.
- Modelled the data in dbt: staging views, an order-lines model, and a star schema (order fact table with customer, product, and date dimensions), with tests on keys and relationships after every load.

ORCHESTRATION AND DEPLOYMENT:

- Ran the full stack locally in Docker Compose with an Apache Airflow DAG (extract, dbt run, dbt test) with retries, and a Metabase dashboard set up from a script.
- Deployed a cloud version where a daily GitHub Actions workflow loads Supabase Postgres and runs the dbt tests.

KEY FEATURES:

- Live Streamlit dashboard with revenue and order trends, top products, and customer lifetime value, refreshed daily.
- Runs on generated sample data (about 2,000 customers, 300 products, and 8,000 orders).

EMR AND TELEHEALTH PLATFORM FOR PATIENT CARE OPERATIONS| CHENNAI |

System Technology Used: MERN (MongoDB, Express.js, React, Node.js)

FRONTEND INTEGRATION:

- Collaborated with the frontend team to integrate backend APIs into a React-based interface for patients, clinicians, and administrators.
- Delivered responsive user interfaces for appointment booking, patient check-in, EMR access, and telehealth session workflows.

BACKEND DEVELOPMENT:

- Built and maintained RESTful APIs using Node.js and Express.js to support patient management, appointments, EMR records, and teleconsultations.
- Designed and optimized MongoDB schemas for patients, providers, appointments, medical records, and audit logs to ensure performance and scalability.
- Implemented backend logic for appointment scheduling, time-slot validation, clinical data management, and role-based access control.

KEY FEATURES:

- Patient self check-in and appointment management with real-time status updates.
- EMR management with secure storage of patient demographics, clinical notes, and visit history.
- Telehealth integration enables virtual consultations between patients and providers.
- Activity logging and audit trails for administrative and clinical actions to support compliance.

CUSTOMER RELATIONSHIP MANAGEMENT (CRM) | CHENNAI |

System Technology Used: MERN (MongoDB, Express.js, React, Node.js)

FRONTEND INTEGRATION:

- Integrated APIs into a React front for seamless user interaction and efficient data visualization.
- Implemented dynamic role-based access control to render components and menus conditionally.
- Collaborated with front-end developers to ensure smooth data flow and enhance user experience.

BACKEND DEVELOPMENT:

- Designed and implemented RESTful APIs using Node.js and Express.js to handle user authentication, lead management, and role-based access control.
- Integrated MongoDB as the database for storing leads, user roles, and permissions with efficient query optimization.
- Developed backend logic for filtering leads based on user roles (Admin, Employee, Tele-counsellor, Super Admin).
- Automated email notifications for reminders and contact form submissions using SMTP.
- Created cron jobs for scheduled tasks like lead follow-up reminders and inactive status updates.

KEY FEATURES:

- Bulk data upload via CSV files with validation for duplicate and missing entries.
- Invoice number auto-generation and lead tracking with detailed activity log.

E-COMMERCE PLATFORM FOR CRACKERS WITH ADMIN PANEL | CHENNAI |

System Technology Used: MERN (MongoDB, Express.js, React, Node.js)

FRONTEND INTEGRATION:

- Collaborated with the frontend team to integrate APIs into a React-based interface for customers and admins.
- Delivered dynamic product listings, search filters, and category-based navigation for customers.

BACKEND DEVELOPMENT:

- Built and maintained RESTful APIs using Node.js and Express.js to support product management, order processing, and user authentication.
- Designed a robust database schema in MongoDB for products, categories, orders, and user data, optimizing queries for performance.
- Created backend logic for calculating discounts, managing inventory, and processing refunds.

KEY FEATURES:

- Dynamic product listings with category-based navigation and search filters.
- Discount calculation on orders, inventory management, and refund processing.

TECHNICAL SKILLS

- **Frontend Development:** HTML5, CSS3, Bootstrap, Tailwind CSS, JavaScript, React.js
- **Backend Development:** Node.js, TypeScript, Express.js, RESTful API Design, Middleware Development, JWT & OAuth-based Authentication, Role-Based Access Control (RBAC)
- **Database Management:** MongoDB, MySQL, SQL Query Writing, Schema Design, Query & Indexing Optimization
- **Data Engineering:** Python, Apache Spark, PySpark, Hadoop, HDFS, Hive, Apache Airflow, dbt, ETL / ELT Pipelines, Data Warehousing, SQL for Analytics
- **Deployment & DevOps:** Docker, AWS EC2, AWS S3, Azure, Linux Server Administration, Nginx, Environment Configuration, GitHub Actions
- **Collaboration:** Git, GitHub
- **Development Tools:** VS Code (Code Editor)
- **Communication Tools:** Slack
- **API Development:** Postman
- **Project Management Tools:** Click Up

ADDITIONAL SKILLS

- Responsive Web Design
- Component-Based Architecture (React)
- RESTful APIs Integration
- State Management (React Context, Hook)
- Command Line Interface (CLI) usage for Git, Node.js, and Linux
- Cron Jobs & Scheduled Task Automation

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I am fully responsible for the authenticity of the details provided and am willing to provide any further information required.